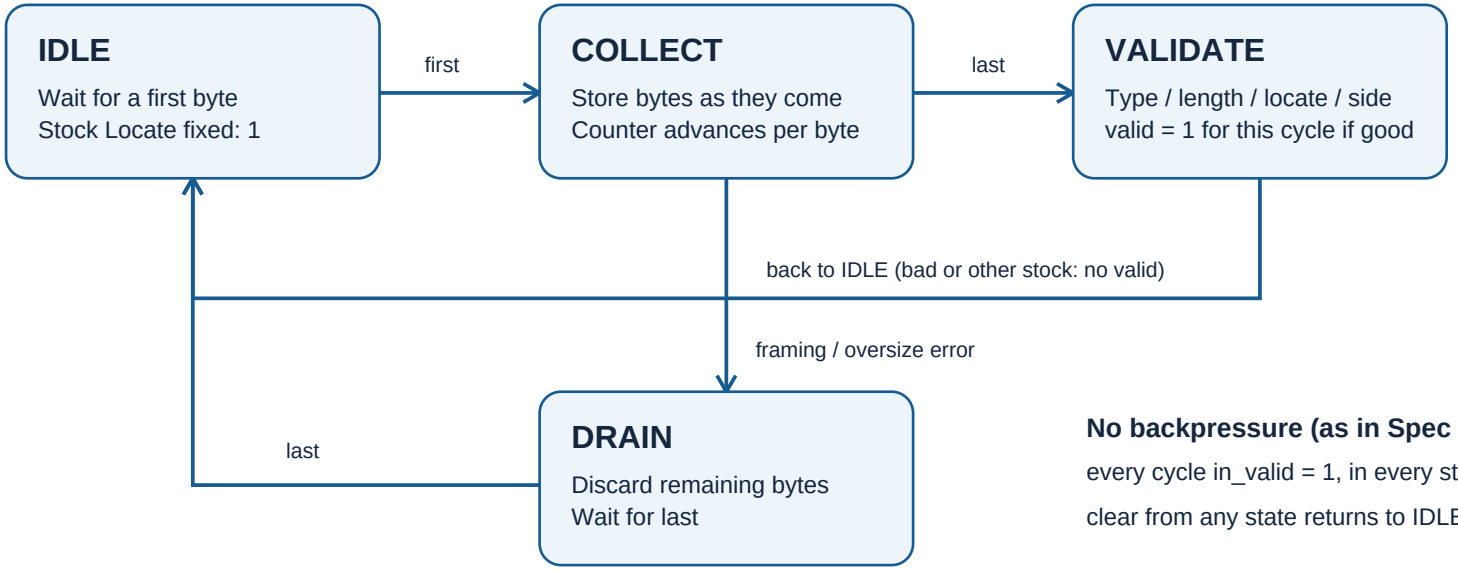
ITCH parser - state-machine sketch

Amirlan / CompE | November 8, 2026 | Implemented decoder

Input: the UDP payload (MoldUDP64 packets), one byte per cycle. A deframer marks each message. Outputs match Spec v1.



Outputs (Spec v1)

valid: one cycle per message
msg_type[1:0]: 0 A, 1 E, 2 X, 3 D
order_ref[63:0]: full wire ref
side: 0 buy, 1 sell (A only)
shares[31:0], price[31:0]

No backpressure (as in Spec v1): a byte is taken

every cycle in_valid = 1, in every state.

clear from any state returns to IDLE on the clock edge.

Boundary cases

First and last together: IDLE -> VALIDATE. Missing first: IDLE -> DRAIN (or IDLE if also last).

In COLLECT, a second first or a 37th byte rejects the frame. If that byte is also last, return directly to IDLE.

Unsupported type, wrong length, bad side or another stock never pulses valid. A cut-off packet drops its last message.

Worked Add Order

58-byte packet: 20-byte header, length 00 24, then 36 bytes: A / locate 1 / ref 42 / Buy / 100 / 1,502,500.

After the message's last byte at edge L: valid is high for one cycle. Cut-through: no packet buffering.

Scope: hw/itch_parser.ml (decoder) + hw/mold_deframer.ml (MoldUDP64 deframer, cut-through; accuracy NOTE inside).

Full contract and transport FSM: docs/parser-interface.md. Tests: hw/test/test_itch_parser.ml.